AI-Driven Health Analyzer Real-Time Physical and Mental Wellness Optimization using ML

25-26J-225

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology (Reading)

Department of Computing

Sri Lanka Institute of Information Technology Sri Lanka

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AI-Powered Fitness Workout Optimization System

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DECLARATION

I affirm that this is my original work and that, to the best of my knowledge and belief, it does not contain any previously published or written material by anyone else, with the exception of instances in which credit is provided within the text. Nor does it incorporate, without acknowledgement, any previously submitted material for a degree or diploma at any other university or institute of higher learning.

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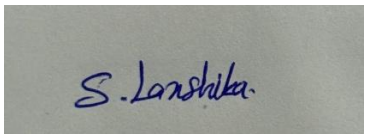
Laxshika S	IT22596802	

Signature of the Supervisor
(Dr. Kapila Dissanayaka)

Date

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ABSTRACT

AI-Powered Workout Optimization System focuses on the development of an AI-powered workout optimization system that uses pose estimation and movement analysis to provide real-time, personalized feedback. By integrating advanced machine learning models and computer vision techniques, the system analyses video input to detect incorrect postures and deliver corrective feedback. The use of supervised learning allows the system to refine its feedback based on labelled data of proper form, ensuring accurate guidance for users across different fitness levels. The system adapts to user progress, continuously personalizing workout suggestions to optimize training and improve performance. As users perform exercises, the system detects posture errors and provides immediate corrections, thereby reducing the risk of injury and improving workout effectiveness. The feedback evolves with each workout session, ensuring that the guidance becomes more precise over time.

By combining AI, machine learning, and computer vision with backend development, this project creates an intelligent fitness coaching system that is responsive and personalized. It represents a significant step forward in AI-driven fitness technologies, offering a smarter approach to workout optimization. The system is designed to empower users to improve their form, track progress, and achieve their fitness goals with higher efficiency and reduced risk. Ultimately, this solution aims to transform the fitness industry by offering a scalable, user-centric platform for real-time movement analysis and feedback.

Keywords: AI-powered fitness system, pose estimation, movement analysis, real-time feedback, personalized workout, machine learning, computer vision, supervised learning, posture correction, injury prevention, workout optimization, fitness technology, intelligent coaching, performance improvement, fitness tracking, adaptive feedback, real-time movement analysis.

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1.INRODUCTION

The growing demand for personalized fitness solutions has driven the emergence of AI-powered systems designed to enhance various aspects of user health and fitness. These systems combine machine learning, computer vision, and real-time feedback to offer tailored advice, ensuring that users receive effective guidance across multiple facets of their well-being. This project aims to develop an integrated AI-driven platform that focuses on optimizing fitness routines, managing hydration, and assessing mental health through intelligent technologies. By collecting and analyzing data from various sources such as user activity, environmental conditions, and personal health metrics. This system will provide users with comprehensive recommendations to help them lead healthier lifestyles. The integration of AI ensures that the platform is adaptive, continuously learning from user data to provide more accurate and relevant feedback. Ultimately, the system aims to redefine fitness tracking by offering an all-encompassing health management tool that not only tracks physical activity but also monitors hydration levels and emotional well-being.

The AI-Powered Workout Optimization System, the primary focus of this project, leverages AI-driven pose estimation and movement analysis to deliver real-time feedback on exercise form. Unlike traditional fitness systems that mainly track basic metrics like steps, calories, and heart rate, this module addresses a critical and often overlooked aspect of fitness: proper posture during exercises. Maintaining correct form is essential for maximizing workout effectiveness and minimizing the risk of injury. Incorrect posture can lead to inefficient workouts and, more importantly, increase the likelihood of strain or injury over time. By utilizing computer vision and pose estimation techniques, the system analyses user movements in real time through video input, identifying posture errors and providing immediate corrective feedback. The system captures key body points, such as joints and limbs, and evaluates their alignment with ideal exercise form. For example, when a user performs a squat, the system assesses the alignment of the knees, hips, and torso, offering corrections if the movement deviates from the optimal form. This system is powered by supervised learning, training on large datasets of properly executed exercises. As a result, the system refines its feedback over time, adapting to individual users and becoming more accurate with each session. As users progress in their fitness journey, the system learns from their movements, adjusting its feedback to match their evolving abilities and goals. This ensures that each user receives personalized, actionable guidance tailored to their needs.

The Workout Optimization module is designed to enhance training efficiency, reduce the risk of injury, and help users achieve their fitness goals through continuous, real-time feedback. By offering personalized suggestions that evolve as users improve, the system aims to provide an intelligent and adaptive fitness coaching experience that grows with the user’s fitness level and performance.

1.1 Background and Literature Survey

1.1.1 Background

The fitness industry has experienced significant advancements with the integration of artificial intelligence (AI), machine learning (ML), and computer vision. Traditional fitness tracking systems generally measure basic metrics such as heart rate, steps, and burning calories, but they fail to address the critical component of correct posture during exercise. Poor form not only reduces the efficiency of workouts but also increases the risk of injury, which is a significant concern in physical training.AI-powered systems, particularly those incorporating pose estimation and movement analysis, offer a solution to this problem by providing real-time, personalized feedback. These systems use computer vision to analyze a user's movements and detect incorrect posture or techniques. Through supervised learning, these systems improve over time, offering more accurate feedback as they learn from data and progress with the user's workouts.

This project aims to leverage AI, pose estimation, and machine learning to create a system that detects posture issues and offers corrective feedback, enhancing workout performance and safety. The system will analyze user movements through video input and adapt its suggestions based on personalized data, promoting continuous improvement.

1.1.2 Literature Survey

The use of pose estimation for real-time feedback in fitness applications has grown significantly in recent years. Open Pose, a state-of-the-art pose estimation model, has shown the capability to track human body movements by identifying key body points in real-time. Cao et al. (2017) introduced Open Pose, which serves as a foundation for fitness applications that require accurate tracking of user posture during dynamic exercises like squats, lungs, or push-ups [1]. Other advanced computer vision models have also demonstrated success in tracking movements with high accuracy, proving essential for providing actionable feedback during workouts.

Machine learning, particularly supervised learning, plays a pivotal role in enhancing the effectiveness of real-time feedback systems. By training models with a large dataset of correct exercise forms, supervised learning algorithms can compare the user's movements to labeled examples, detect deviations, and provide corrective suggestions. Gupta and Soni (2021) highlight the potential of AI-driven posture correction systems, emphasizing the accuracy of feedback provided through supervised learning in fitness applications [3]. These systems not only detect posture errors but also personalize feedback over time, adapting to a user's specific needs.

Moreover, AI systems have shown promising results in boosting user engagement and adherence to fitness routines. Lee and Kwon (2021) discuss how AI-powered systems can provide personalized workout plans by analyzing individual performance and recommending adjustments based on the user's progress [4]. Real-time corrective feedback encourages users to maintain proper form and avoid injury, as outlined by Hassani and Khojasteh (2020) [5]. Furthermore, AI-based systems have been instrumental in enhancing motivation by providing users with

immediate feedback and progress tracking. This engagement factor has been proven to increase workout adherence, as individuals feel more connected to their training goals. AI applications not only focus on physical form correction but also contribute to long-term fitness by continuously adapting workout plans, thus ensuring that the system remains relevant to the user's evolving capabilities.

In summary, the literature supports the integration of AI, pose estimation, and machine learning for creating intelligent fitness systems. These systems provide real-time, personalized feedback that enhances the effectiveness of workouts, reduces the risk of injury, and promotes long-term fitness progress through continuous adaptation.

1.1.3 Survey Response

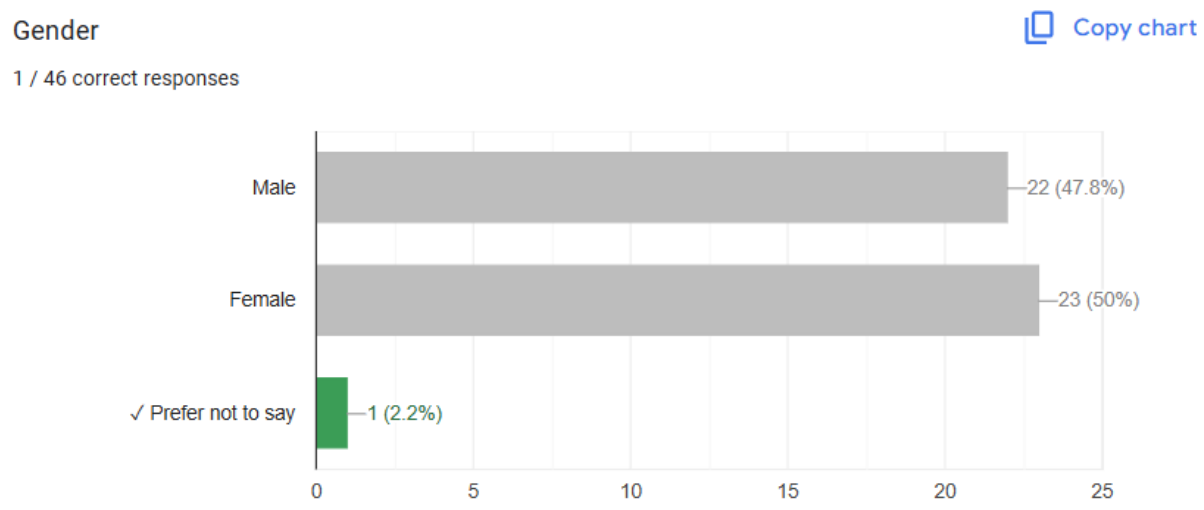

Figure 1: Survey 1

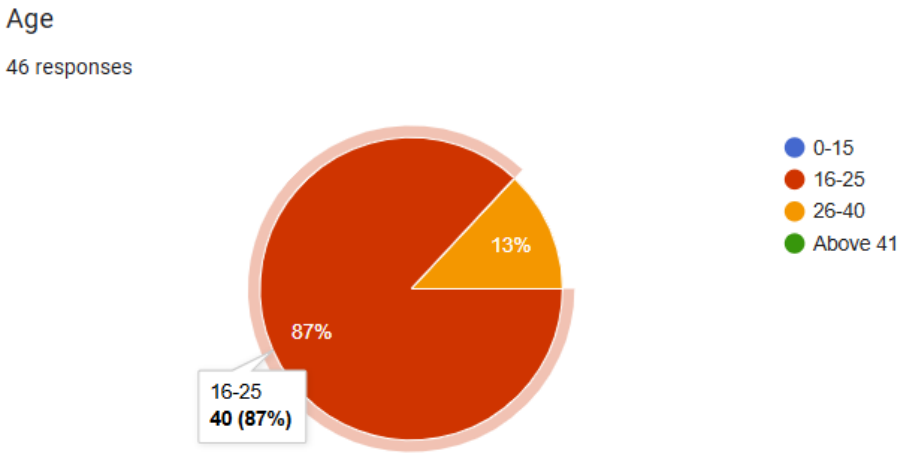

Figure 2 : Survey2

How often do you engage in physical workouts?

46 responses

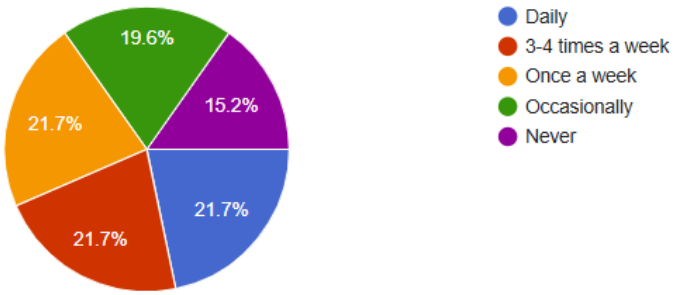

Figure 3 : survey physical workout

How important is posture correction to you during workouts?

46 responses

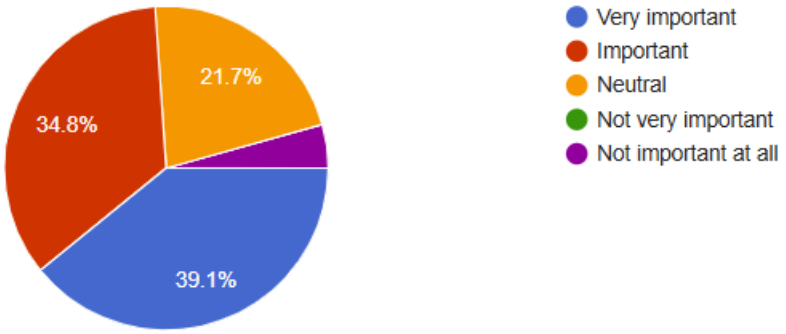

Figure 4 : need

Have you ever used any fitness tracking app or wearable device that provides feedback on exercise posture?

46 responses

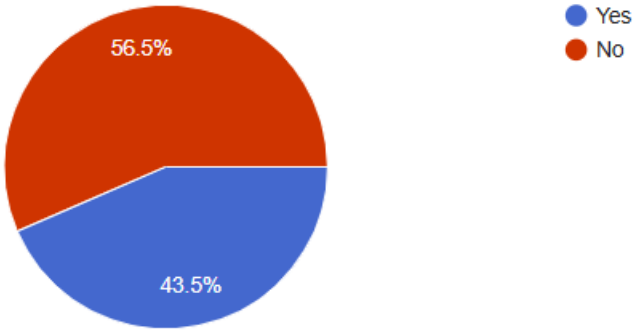

Figure 5 :used app

How comfortable are you with using AI-based fitness technologies for workout guidance?

46 responses

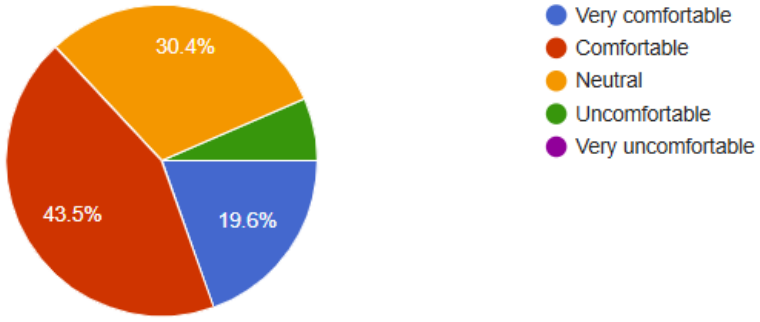

Figure 6 : how comfortable

How likely are you to use a system that provides real-time feedback on your exercise form and posture?

46 responses

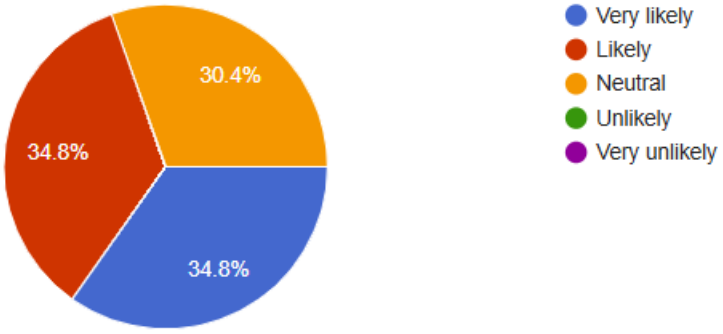

Figure 7 : how much like use

Do you think that AI-based feedback during workouts would help reduce the risk of injury?

46 responses

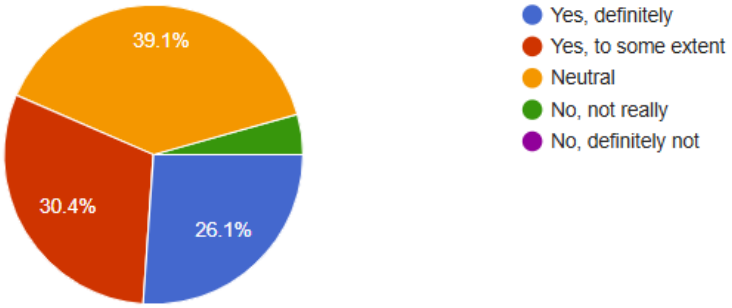

Figure 8 : reduce the injury

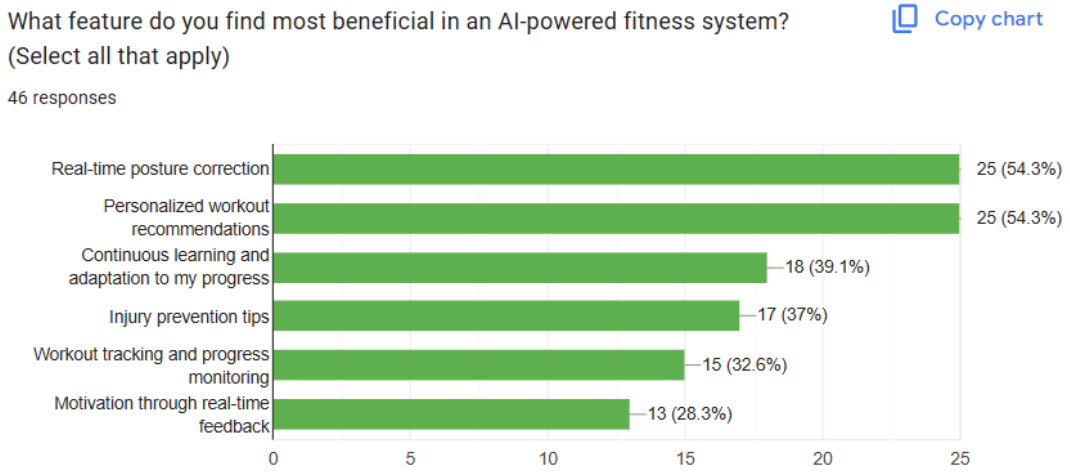

Figure 9 : features

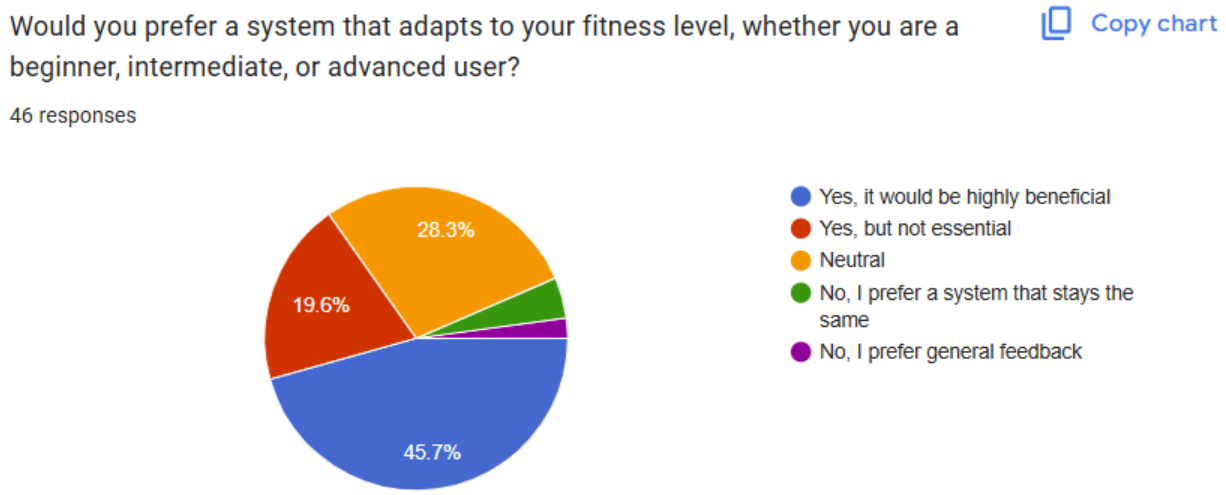

Figure 10 :fitness level

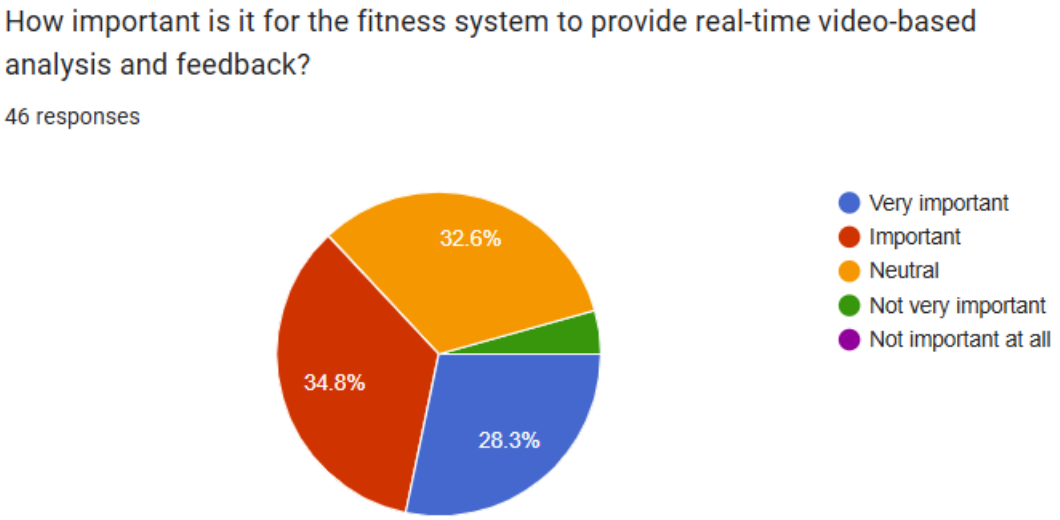

Figure 11 : video based

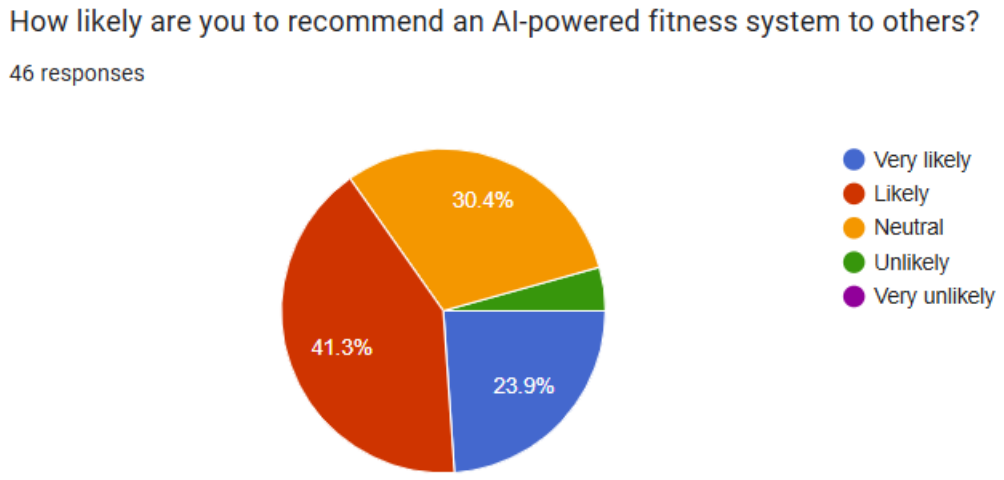

Figure 12: recommended

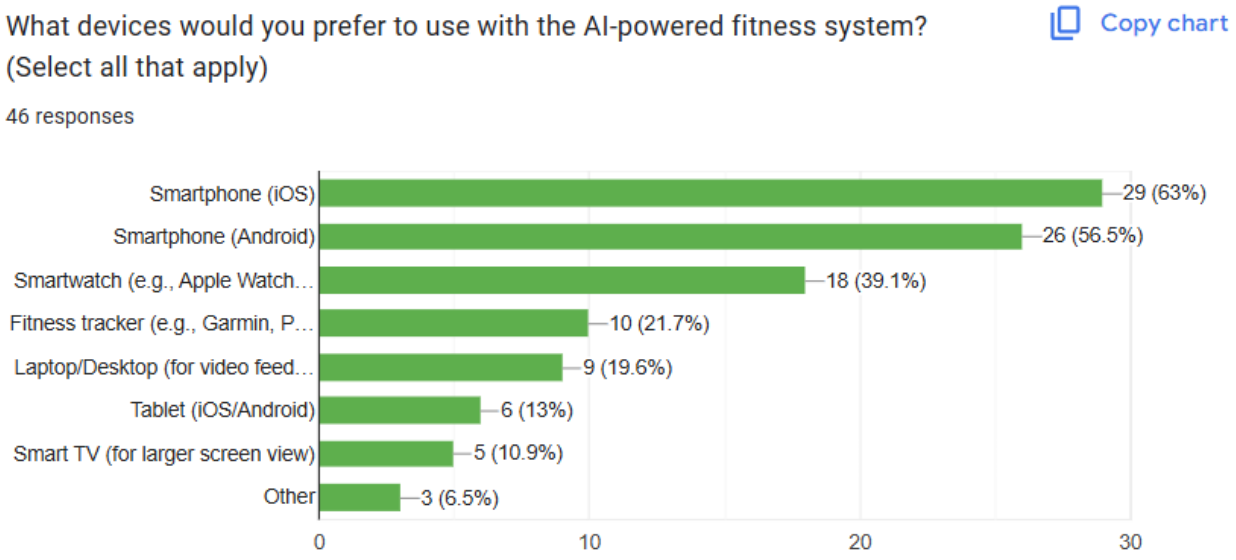

Figure 13: device

1. 2 Research Gap

While significant progress has been made in the integration of AI and machine learning for fitness applications, several gaps remain that limit the potential of these technologies in real-time workout optimization. First, many existing systems provide feedback on general performance metrics but fail to offer immediate corrective feedback on posture during dynamic exercises. Real-time posture correction, particularly for complex movements, remains underdeveloped. Second, most AI-powered fitness systems rely on predefined templates or static models for feedback, which do not evolve with the user's progress. There is a distinct lack of personalized, adaptive systems that adjust their feedback based on an individual's performance, requiring systems that can continuously refine their suggestions as the user develops over time.

Another gap is the limited incorporation of continuous learning from user interactions. While some systems use machine learning, they often do not fully leverage user data over time to adapt and enhance feedback. A more robust system that learns from each user's unique movement patterns is essential for long-term engagement and improvement. Additionally, while many fitness systems attempt to correct posture, they often overlook real-time injury prevention. Real-time feedback that identifies and prevents posture mistakes that could lead to injury is an area that still requires significant development, especially for high-impact exercises. Lastly, existing systems tend to cater to users at one level of fitness, often neglecting scalability. A key challenge lies in creating systems that can adapt to varying fitness levels, from beginners to advanced athletes, ensuring that the feedback and recommendations remain effective across a broad spectrum of users. This research aims to address these gaps by developing a comprehensive AI-driven system that provides real-time feedback, personalizes coaching based on user progress, focuses on injury prevention, and adapts to users' fitness levels.

Table 1: Comparison of Previous Research

Research	Real time Posture Correction	Personalized Feedback	Injury prevention	Adaptability Across Fitness Levels	Real Time Video Analysis
Cao et al. (2017) [1]	No	No	No	No	No
Gupta & Soni (2021) [3]	Yes	No	No	No	Yes
Lee & Kwon (2021) [4]	No	Yes	No	Yes	No
Hassani & Khojasteh (2020) [5]	Yes	No	Yes	No	Yes
Zhang et al. (2020) [2]	Yes	Yes	Yes	Yes	No
My Project	Yes	Yes	Yes	Yes	Yes

Table 2 : Problems and Outputs

Problem Addressed	**Technique**	**Analytical Output Parameters**
Real-Time Posture Correction	Pose Estimation, Supervised Learning	Detection of posture errors, real-time feedback, corrective suggestions
Personalization of Feedback Based on User Progress	Supervised Learning, Continuous Learning	Adaptation of feedback over time, personalized coaching adjustments
Injury Prevention During Exercise	Pose Estimation, Machine Learning	Detection of posture faults that may lead to injury, preventive tips
Feedback Adaptation for Different Fitness Levels	AI-driven Fitness Models	Tailored recommendations based on skill level, workout intensity adjustment
Lack of Integration Between Pose Estimation and Real-Time Feedback	AI and Machine Learning with Video Input	Real-time posture and movement analysis, corrective suggestions based on real-time video input

1.3 Research Problem

The fitness industry has increasingly relied on technology to provide personalized experiences for users, ranging from basic fitness tracking to advanced AI-driven coaching. Despite these advancements, there is still a significant gap in addressing real-time feedback on exercise form, which is crucial for improving workout efficiency and preventing injury. Most existing systems focus on tracking basic metrics such as calories burned or steps taken but fail to correct exercise posture in real time or adapt to individual user progress.

Incorrect posture during exercise is a major contributor to both reduced workout effectiveness and the risk of injury. Traditional fitness apps or devices primarily provide generic feedback based on pre-set models, which are not personalized to the user’s body mechanics or progress. This lack of real-time, personalized feedback limits the potential for users to improve their form and maximize their workout results while maintaining safety. Furthermore, many of these systems fail to continuously learn from user interactions, meaning their feedback does not evolve with the user's growth or changing fitness levels.

The challenge lies in developing an intelligent system that can analyze body movements in real-time using pose estimation and machine learning, providing immediate feedback for posture correction. Additionally, such a system must adapt over time to a user's individual progress, offering personal recommendations that improve with each session. Furthermore, there is a need for continuous learning models that refine the feedback based on historical data, creating a dynamic and responsive coaching experience.

This research seeks to address these issues by developing an AI-powered system that uses real-time video input, pose estimation, and machine learning to provide personalized, adaptive feedback for optimizing exercise form, preventing injuries, and enhancing long-term fitness progress. By integrating continuous learning, this system aims to bridge the gap between traditional fitness tracking tools and advanced, real-time workout optimization.

Table 3 : Research Descriptions

Research Element	Description
Research Problem	Current fitness systems lack real-time, personalized feedback on posture, and do not adapt based on user progress or continuous learning.
Focus on Existing Research	Research focuses on pose estimation, machine learning for personalized fitness, and injury prevention but lacks real-time adaptation.
Identified Gap	Existing systems do not integrate real-time video feedback, personalized adjustments, or continuous learning from user data.
Extended Research Problem	Developing an AI system that combines pose estimation, real-time feedback, and continuous learning to optimize posture and prevent injury.
Key Ethical Concerns	Concerns about user data privacy, video analysis, and ensuring secure storage of personal data.
Proposed Solution	AI-powered system offering real-time personalized feedback and continuous adaptation to improve posture and prevent injuries
Objective	To create a system that provides adaptive feedback for workout optimization and injury prevention.
Contribution	Providing a more intelligent and adaptive fitness solution that bridges the gap between current tracking tools and AI-powered coaching.

2.Objectives of the Project

2.1 MAIN OBJECTIVE

The project aims to develop an AI-Driven Health Analyzer for Real-Time Physical and Mental Wellness Optimization using Machine Learning. The system will provide personalized feedback on physical fitness, mental health, and hydration, helping users optimize their wellness through real-time insights and recommendations. By integrating AI, machine learning, and computer vision, it will enable users to improve exercise routines, manage hydration, and enhance emotional well-being.

The AI-Powered Fitness Workout Optimization System, a core component, focuses on providing real-time feedback on exercise form using pose estimation and supervised learning. The system analyses user movements, detects posture errors, and offers corrective feedback to improve workout efficiency, reduce injury risks, and help users achieve their fitness goals.

2.2 Specific Objectives

2.2.1 Data Collection and Storage

Collect real-time video data of users performing exercises using cameras or webcams. Store the data securely in a database while maintaining user privacy and ensuring data integrity

2.2.2 Data Preprocessing and Feature Engineering

Process raw video input to extract relevant features such as joint positions, angles, and movement trajectories. Normalize and label the data to create a high-quality dataset suitable for supervised learning.

2.2.3 Machine Learning Model Development

Develop and train AI models for pose estimation, movement analysis, and posture evaluation. Implement supervised learning algorithms that compare user movements against labeled examples of correct exercise forms.

2.2.4 Demographic Categorization

Incorporate user demographic information (age, fitness level, and physical attributes) to provide personalized guidance and ensure exercises are appropriate and safe.

2.2.5 Personalized Workout Plan Generation

Generate adaptive workout recommendations based on the user's progress, fitness level, and exercise history. Continuously refine the plan with real-time feedback from the AI system.

2.2.6 Testing and Validation

Evaluate the accuracy and performance of the pose estimation and feedback system using benchmark datasets and user trials. Ensure the system reliably detects incorrect postures and provides meaningful corrections.

2.2.7 Ethical Considerations

Ensure privacy, consent, and data security. Prevent misuse of personal video data and maintain ethical AI practices during system development and deployment.

2.2.8 Integration and Deployment

Integrate AI models with a responsive user interface for real-time interaction. Deploy the system on suitable platforms (mobile, desktop, or web) for user accessibility.

2.2.9 User Feedback and Iterative Refinement

Incorporate user feedback to iteratively improve system performance, personalization, and usability. Adapt recommendations based on user engagement and results.

2.2.10 Evaluation of Ethical Impact:

Assess the impact of the system on user privacy, safety, and overall trust. Ensure compliance with ethical standards in AI-driven fitness applications.

3. METHODOLOGY

3.1 Project Proposed Execution

The execution of the **AI-Powered Fitness Workout Optimization System** involves several stages, from data collection to deployment. Below are the key tools and technologies that will be utilized to develop and deploy the system

Tools and Technologies for AI-Powered Fitness Workout Optimization System

1. Machine Learning Frameworks:
    - TensorFlow: A powerful open-source framework for machine learning and deep learning, used for developing models related to pose estimation and movement analysis.
    - PyTorch: A flexible deep learning framework for building and training models. It's commonly used for research and experimentation, particularly for computer vision tasks.
    - Scikit-Learn: A machine learning library used for preprocessing data, model training, and evaluation, especially for supervised learning models like decision trees and regression.
2. Pose Estimation & Computer Vision:
    - OpenPose: A widely-used library for real-time multi-person keypoint detection, essential for pose estimation. It helps track joint locations in real-time during exercises.
    - MediaPipe: A framework by Google for building cross-platform AI solutions, including pose estimation and gesture recognition, ideal for real-time workout analysis.
    - OpenCV: A library for computer vision tasks, used to process video frames and handle visual inputs for real-time feedback generation.
3. Backend Development:
    - Flask/Django (Python): Lightweight backend frameworks used to handle user interactions, integrate machine learning models, and manage API requests for real-time feedback.
    - Node.js: Used for scalable server-side processing, especially if the system will have a web-based interface for real-time feedback and interaction.
4. Database:

- MongoDB: A NoSQL database that will be used to store user profiles, workout history, feedback data, and progress logs. MongoDB is scalable and flexible, ideal for storing diverse user data.
- PostgreSQL/MySQL: Relational databases that could be used for structured data storage, especially if relational querying is needed for advanced features.

5. Frontend Development:
    - React: A JavaScript library for building user interfaces, ensuring that the system is responsive, interactive, and intuitive. It will be used to build dashboards, feedback displays, and other UI components.
    - Redux: A state management library for React, which helps in managing the state across different components, ensuring smooth user experience across real-time interactions.
6. Deployment:
    - Docker: Containerization technology that allows for packaging the application with all its dependencies, ensuring it runs consistently across different environments.
    - Kubernetes: For orchestrating the deployment of the system, especially in cloud environments, managing scalability, and ensuring high availability.
    - AWS/GCP: Cloud platforms that will host the system, enabling scaling and flexibility in processing power, storage, and machine learning model execution.
7. Version Control:
    - Git: A distributed version control system used to manage the codebase, track changes, and collaborate with the development team.
    - GitHub/GitLab: Platforms for hosting the Git repository, facilitating code collaboration and CI/CD integration.
8. Real-Time Video Processing:
    - WebRTC: A real-time communication protocol used to handle live video input for pose estimation during workouts.
    - FFmpeg: A multimedia framework used to process video and audio data, crucial for real-time video stream handling.
9. Security and Privacy:
    - OAuth: An authentication framework for managing user identities securely and ensuring authorized access to the system.
    - SSL/TLS: For encrypting communications between the user interface, backend servers, and databases to maintain data privacy.

3.2 Complete System Architecture

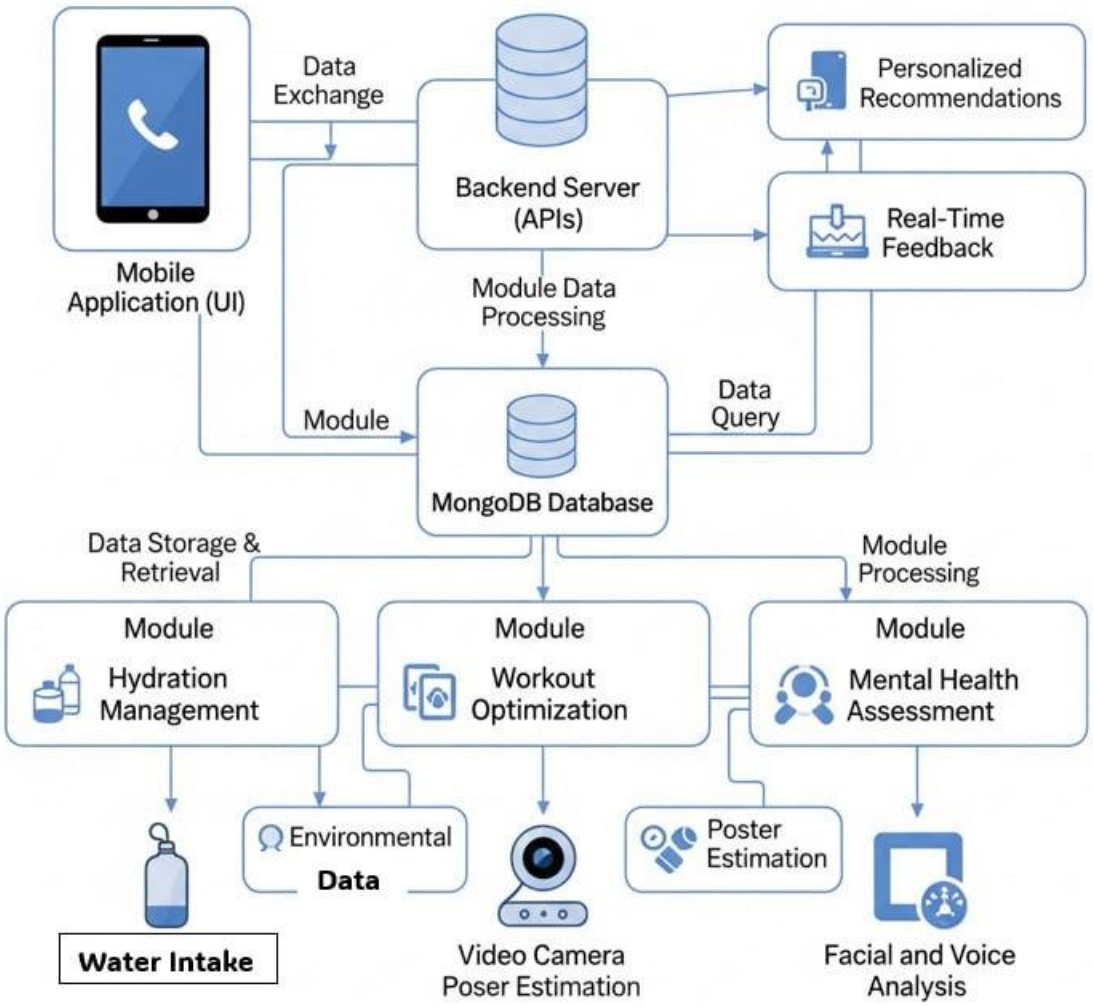

Figure 14 : System Diagram

3.3 Component Architecture

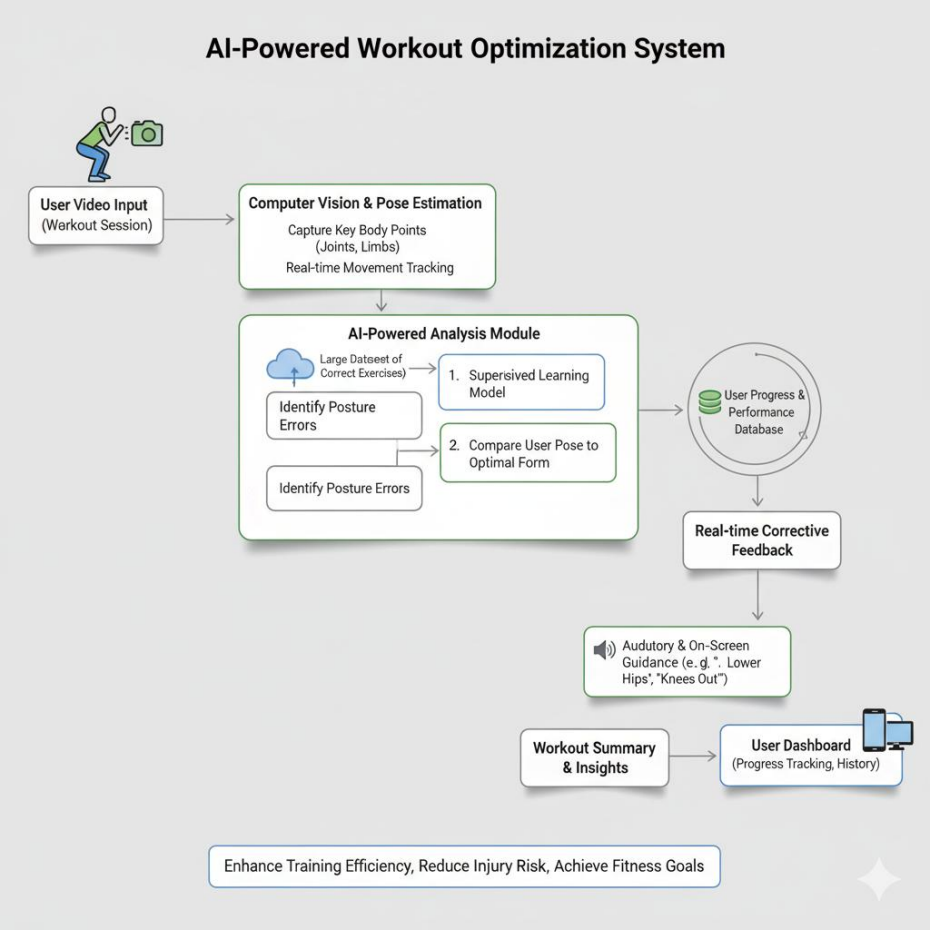

Figure 15 : Component Architecture

3.4 Tasks and Sub Tasks

Task 1-Data Collection
Gather video data from users performing various exercises.
Collect demographic information (e.g., fitness level, age) to personalize the system.
Task 2-Data Preprocessing and Feature Engineering
Extract joint positions and body movements from video frames.
Normalize and label data for training.
Task 3-Machine Learning Model Development
Develop AI models for pose estimation.
Train supervised learning models for posture correction.
Task 4-Testing and Validation
Test the model's accuracy in real-world scenarios.
Validate the system with benchmark datasets.
Task 5-Integration and Deployment
Build the user interface and integrate AI models.
Deploy the system for real-time feedback on mobile and desktop platforms.

Task 6-User Feedback and Iterative Refinement
Collect feedback from early users.
Continuously improve the system based on user interactions.

3.5 Required Materials

Hardware:

- Camera/Webcam for video input
- Computer (desktop/laptop) or mobile device for processing
- A server for storing and processing data securely

Software:

- Machine learning frameworks (TensorFlow, PyTorch)
- Pose estimation algorithms (OpenPose or similar)
- Backend technologies for deployment (Node.js, Django, etc.)
- Database systems for data storage (MySQL, PostgreSQL, etc.)

Tools:

- Video editing software for preprocessing (if needed)
- IDEs for development (Visual Studio Code, PyCharm, etc.)

3.4 Time Frame

The project will be executed over a 12-month period, broken into four major phases:

- **Phase 1: Data Collection & Preprocessing (2 months)**
  - Gather user data, preprocess video inputs, and build initial datasets.
- **Phase 2: Model Development & Testing (4 months)**
  - Develop and train machine learning models for pose estimation and feedback generation.
- **Phase 3: Integration & Deployment (3 months)**
  - Integrate the models with a user interface and deploy the system.
- **Phase 4: User Testing & Refinement (3 months)**
  - Conduct real-world user testing and refine the system based on feedback.

3.6 Schedules

Task Charts: Gantt charts will be used to visualize the task dependencies and timelines, ensuring a structured and coordinated workflow throughout the project. This will help track the progress of various phases, from data collection to deployment and refinement.

Team Collaboration: The project team will be divided into specialized roles to ensure focused development. Frontend/UI Developer: Responsible for designing and implementing the user interface and experience. Backend Developer: Handles server-side logic, API integration, and system deployment. Machine Learning Specialist: Develops and trains AI models for pose estimation, movement analysis, and feedback generation. Database Manager: Manages data collection, storage, and preprocessing, ensuring security and privacy.

3.7 Anticipated Conclusion

Results: The project aims to produce a highly accurate fitness system that combines AI-driven pose estimation with machine learning to offer personalized workout feedback and suggestions. The system will adapt to user data, ensuring it evolves with individual progress, providing actionable insights to improve posture, performance, and safety during workouts.

Application: The developed system will function as an advanced, AI-powered personal trainer, offering real-time feedback and generating customized workout plans tailored to individual health levels, fitness goals, and preferences.

Real-World Use: The final system, deployed as a web-based application, is expected to be a valuable resource for users seeking personalized fitness guidance. Its accessibility and personalized nature will make it an impactful tool for individuals looking to optimize their workout routines and improve overall health and well-being.

4.PROJECT REQUIRMENTS

4.1 Functional Requirements

Real-Time Feedback on Posture

The system must provide real-time feedback on the user's posture and form during workouts by analysing video inputs using pose estimation models.

Pose Estimation and Movement Analysis

The system must accurately estimate human body poses and analyse movement to detect posture errors during exercises.

Personalized Workout Recommendations

Based on user progress and fitness levels, the system should suggest adaptive workout plans that evolve over time to match the user’s goals and improvement.

Injury Prevention Tips

The system must provide real-time injury prevention recommendations based on posture analysis to reduce the risk of exercise-related injuries.

User Data Tracking

The system should collect and store user performance data, including past workout history, feedback, and progress for future analysis and improvement.

Continuous Learning and Adaptation

The system must learn from user performance and continuously improve feedback, ensuring it adapts to the user’s evolving fitness level.

4.2 Non-Functional Requirements

Scalability-The system should be able to scale to handle multiple users simultaneously, without affecting performance, particularly when deployed in cloud environments.

Security and Privacy-The system must ensure that all user data, including video inputs, is securely stored and processed, with appropriate privacy controls to protect sensitive data.

Usability-The system's user interface must be intuitive and user-friendly, allowing users to easily navigate and interact with the application without requiring technical expertise.

Performance-The system should provide real-time feedback with minimal latency (less than 1 second) to ensure users can act on the feedback immediately.

Cross-Platform Compatibility-The system must be accessible across different platforms (mobile, desktop, and web applications) to cater to various user preferences.

4.3 System Requirements

In the intricate architecture of the AI-powered fitness system, each component is meticulously designed to ensure seamless interaction and performance. The frontend, built using React, serves as the gateway to the user experience, combining aesthetic appeal with intuitive navigation. This interface enables users to effortlessly upload workout videos, view real-time feedback, and track their fitness progress, ensuring a highly engaging and user-friendly environment.

The backend is powered by Flask or Django (Python), providing a robust foundation for processing data and interacting with Machine Learning (ML) models. This core layer is responsible for processing the video data, estimating posture through pose estimation models, and generating personalized workout feedback. The backend also facilitates user profile management and real-time feedback delivery. To ensure scalability and handle user data effectively, the system utilizes MongoDB as its database of choice, chosen for its flexibility in storing user-specific information and workout logs.

The heart of the system lies in its machine learning models, which harness complex algorithms like Radial Basis Function Neural Networks (RBF-NN) and Decision Trees to process user data and deliver accurate, personalized workout plans. These models are essential for posture estimation, providing real-time feedback on exercise form, and continuously adapting to user progress.

Security and privacy are paramount in the system's design. The platform implements advanced encryption and anonymization techniques, ensuring that sensitive user data remains protected in compliance with privacy regulations. Scalability is also a key consideration, as the system is built to efficiently handle a growing user base without compromising performance. The entire system undergoes rigorous testing, from unit tests to end-to-end assessments, ensuring the accuracy and reliability of health predictions and workout recommendations.

Ultimately, this system combines cutting-edge technology with a focus on resilience, privacy, and user satisfaction, creating a comprehensive solution that adapts to the needs of every individual user.

4.4 User Requirements

The AI-powered fitness system must offer an intuitive and seamless user interface, crafted with React, to ensure a smooth experience for users when inputting health-related parameters like weight, height, and blood pressure. Users should be able to provide demographic details such as age, gender, and fitness level to receive tailored workout recommendations.

The system must allow users to set workout preferences, such as type (e.g., cardio, strength training) and intensity (low, medium, high), to ensure that recommendations align with their fitness goals and abilities. Additionally, users should have the option to track their progress, with clear visual representations of improvements and milestones reached.

A key feature is the real-time feedback mechanism, which will provide users with immediate suggestions for improving posture and form during exercises. The system must also allow users to rate their workout experiences, providing valuable feedback for further refinement and personalization of the system's recommendations.

Privacy and data security must be fundamental in the system design. All user data, including health metrics and video inputs, must be securely stored with encryption and anonymization to protect user privacy and meet legal requirements. The system should also provide users with clear consent forms and privacy policies before collecting their personal data.

By ensuring a highly responsive and personalized experience, the system aims to empower users to take control of their fitness journey while ensuring that their data remains safe and confidential.

5.BUDGET AND BUDGET JUSTIFICATION

5.1 Budget

Cloud Services LKR 40,000
MongoDB Atlas LKR 15,000 annual
Advertisement LKR 1,500
Development Tools LKR 2,000

5.2 Budget Justification

1.Cloud Services (LKR 40,000 annually):
The largest portion of the budget is allocated for cloud services like AWS, GCP, or Azure. These services provide essential resources such as compute power (EC2, Lambda), data storage, and other necessary cloud infrastructure to support the AI-driven hydration management system. The cloud services are crucial for scaling the system as user demand increases and for ensuring high availability and performance
2.MongoDB Atlas (LKR 15,000 annually):
MongoDB Atlas is selected for its efficiency in managing large volumes of unstructured data, such as user profiles, hydration history, and environmental data. The annual cost includes database hosting, backups, and security services, ensuring that the database is accessible, secure, and scalable.
3. Advertisement (LKR 1,500/month):
A portion of the budget is dedicated to social media advertising on platforms like Facebook and Instagram. This helps increase awareness of the platform after launch and allows for targeted audience to reach maximize user acquisition. These platforms offer affordable paid post boosting options that will drive traffic to the mobile application or website. 4.Development Tools (LKR 2,000 annual): Minimal expenses are allocated for development tools and software libraries essential for coding and machine learning model development. This includes IDEs, version control services (like GitHub), and other utilities to support the backend and mobile app development.

5.3 Commercialization

1. Subscription Model
Overview: A subscription-based pricing model will be implemented to offer users various pricing tiers.
Basic Plan: Provides essential features, such as basic hydration tracking and general recommendations.
Premium Plan: Offers advanced features, such as personalized hydration plans, daily recommendations based on detailed activity logs, and in-depth health analytics.
2. Revenue Generation: Monthly or yearly subscriptions will generate recurring revenue, ensuring consistent cash flow and incentivizing users to stay engaged with the platform over time

6. Work breakdown structure

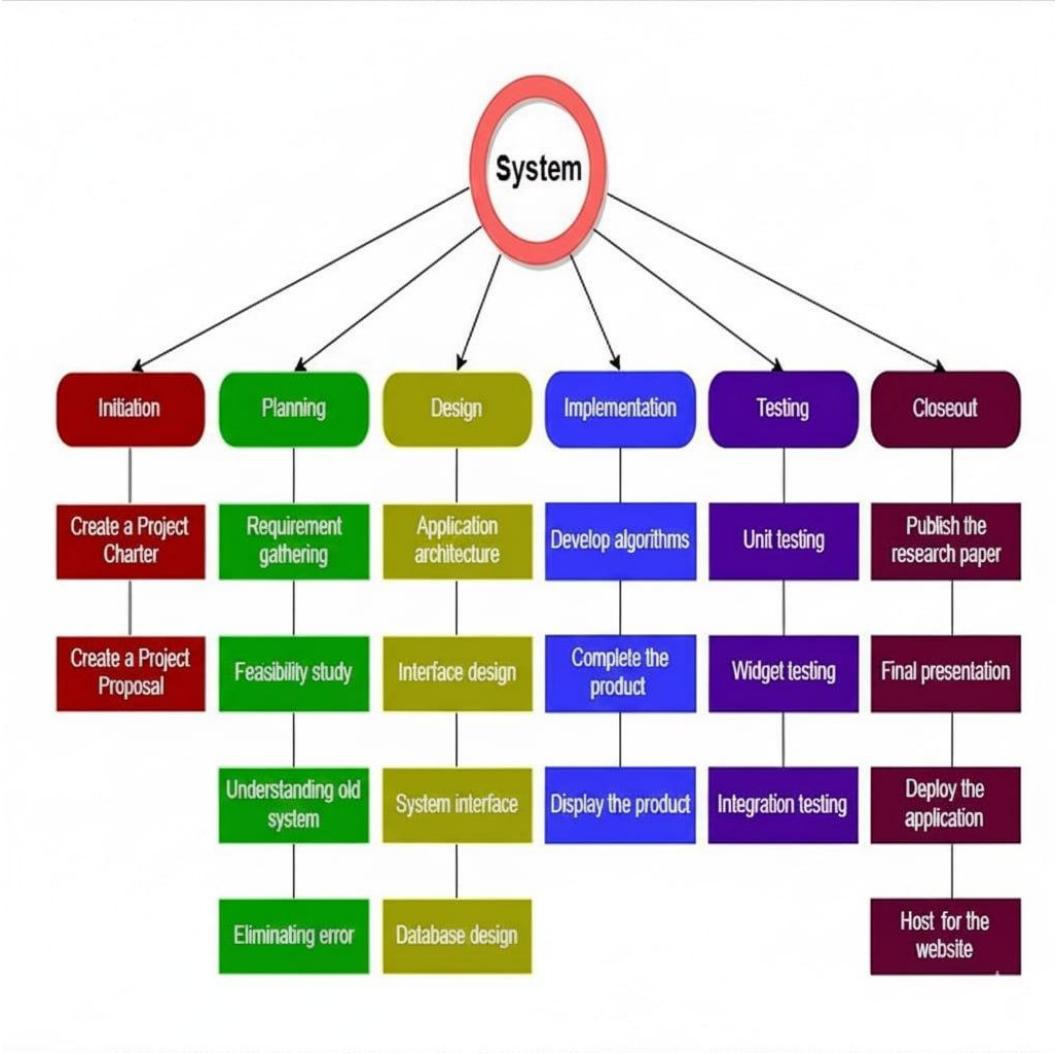

Figure 16 : work breakdown structure

7. Gantt Chart

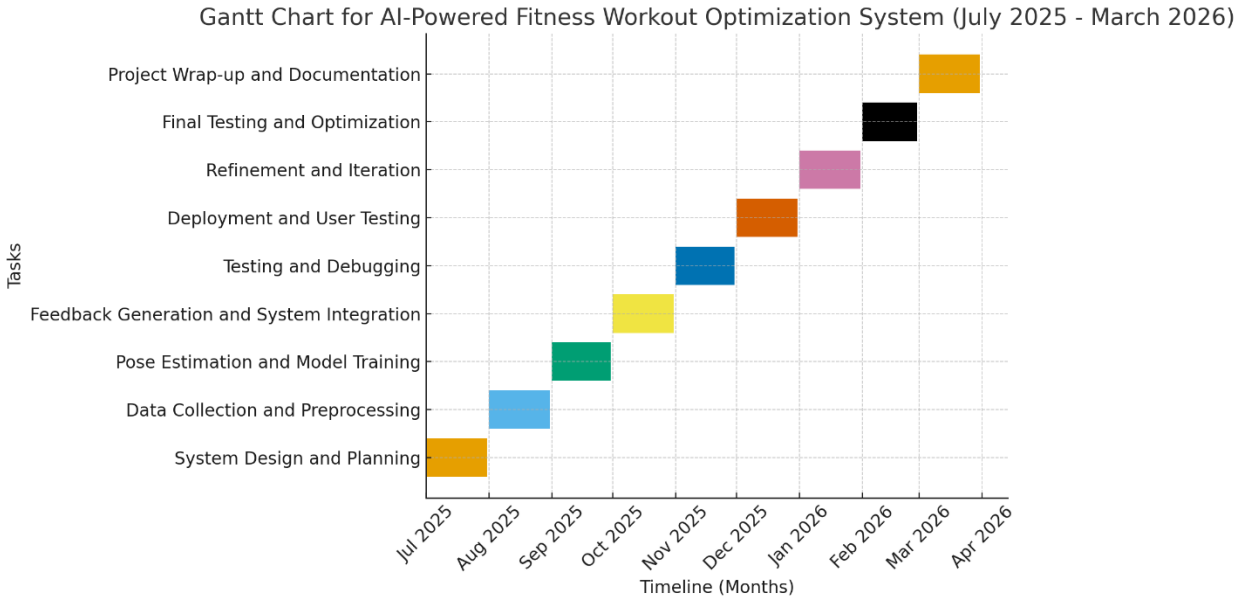

Figure 17 : Gann Chart

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7.Appendices

Appendix A - Survey Link

https://docs.google.com/forms/d/e/1FAIpQLSddcnb8lCXVchkI8GwyEEr_tDEy_CL9kEQONRE9TCjQTw2nkA/viewform?usp=sharing&ouid=116087803700371539290